TA Session 3 - Fast Paced

Questions

Discrete Random Variables

1. Five men and 5 women are ranked according to their scores on an exam (no ties, all 10! rankings equally likely). Let X denote the highest ranking achieved by a woman (for instance, $X = 1$ if the top-ranked person is female). Find $P\{X = i\}$ for $i = 1, 2, \ldots, 6$. *Source: Chapter 4, End-of-Chapter Problems, Problem 1, p. 142*

$P(X=1)$ - Highest ranking by woman = 1

Rankings are equally likely.

$$P(X=1) = \frac{\text{No. of women}}{\text{Total No. of people}} = \frac{5}{10} = \frac{1}{2}$$

$$P(X=2) = P(\text{Rank 1 is a man}) \cdot P(\text{Rank 2 is a woman})$$

$$\frac{5}{10} \times \frac{5}{9} = \frac{5}{18}$$

$\frac{5}{9}$ → Prob. a woman rank 2

4 men

5 women

5 men + 5 women

$$P(X=3) = 5/10 \times 4/9 \times 5/8$$

$$P(X=4) = 5/10 \times 4/9 \times 3/8 \times 5/7$$

2. Let X represent the difference between the number of heads and the number of tails obtained when a fair coin is tossed n times. What are the possible values of X? *Source: Chapter 4, End-of-Chapter Problems, Problem 2, p. 142*

Coin tossed n times

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No. of heads = H

No. of tails = n − H

What to find?

Heads − Tails = $H - (n - H)$

$= 2H - n$

0 heads - - - . . . n heads

The values taken by X

$-n \quad 2-n \quad \ldots \quad 2n \cdot n$

X can take values from $(-n \ldots n)$ with a gap of 2

$-n \quad -n+2 \quad \ldots \quad n$

Ans: X can takes values from $-n \ldots n$ gap of 2

$-n, \quad -n+2 \quad \ldots \quad n$

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X = No. of heads − No. of tails

3. In the previous problem, if the coin is fair and $n = 3$, what are the probabilities associated with each value that X can take on? *Source: Chapter 4, End-of-Chapter Problems, Problem 3, p. 142*

Ω = {HHH, HHT, HTH, THH, TTH, THT, HTT, TTT}

$n = 3$

$X = -3 \ldots\ldots 3$

$X = -3, -1, 1, 3$

$P(X=-3) \rightarrow P(\text{All tails}) = P(TTT) = \frac{1}{8}$

No. of heads − Tails $= -3$

$\Rightarrow$ 3 tails and 0 head.

$X \sim Bin(n, p)$

$Bin(3, 1/2)$

$P(X=-1) = P(2 \text{ Tails}) = 3/8$

H − T = −1

1 − 2

$P(X=1) = 3/8$

$P(X=3) = 1/8$

Probability Mass Functions

4. The distribution function of the random variable X is given by

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{2}, & 0 \le x < 1, \\ \frac{2}{3}, & 1 \le x < 2, \\ \frac{11}{12}, & 2 \le x < 3, \\ 1, & 3 \le x. \end{cases}$$

$P(X<0)$

$P(X<1)$

$P(X<2)$

$P(X<3)$

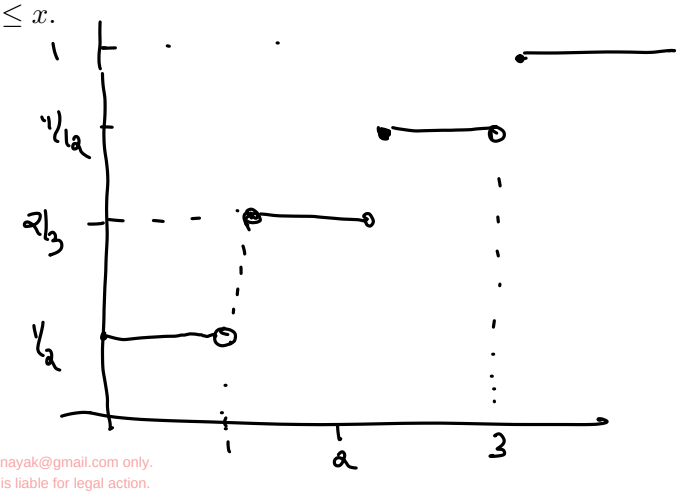
(a) Plot this distribution function.

(b) What is $P\{X > \frac{1}{2}\}$?

(c) What is $P\{2 < X \le 4\}$?

(d) What is $P\{X < 3\}$?

(e) What is $P\{X = 1\}$?



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$P(X > 1/2) = 1 - P(X \leq 1/2)$

$= 1 - 1/2 = 1/2$

$P(2 < X \leq 4) = F(4) - F(2)$

$1 - \frac{11}{12} = 1/12$

$P(X < 3) = F(3^-)$

$= \frac{11}{12}$

$P(X = 1) = F(1) - F(1^-)$

$= 1/6$

$P(X \leq 1)$

$= P(X < 1) + P(X = 1)$

$P(X = 1) = P(X \leq 1) - P(X < 1)$

$F(1) - F(1^-)$

$\frac{2}{3} - 1/2$

$= 1/6$

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Source: Chapter 4, End-of-Chapter Problems, Problem 4, p. 142

5. The probability mass function of a random variable X is given by $p(i) = c\lambda^i/i!$, $i = 0, 1, 2, \ldots$, where λ is some positive value. Find (a) $P\{X = 0\}$ and (b) $P\{X > 2\}$.

$$p(i) = c\frac{\lambda^i}{i!}$$

$$\sum_{i=0}^{\infty} p(i) = 1 \quad = \quad c \cdot \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

$$c \cdot e^{\lambda} = 1$$

$$c = e^{-\lambda}$$

$$P(X = i) = e^{-\lambda}\frac{\lambda^i}{i!} \qquad X \sim \text{Poisson}(\lambda = 0.1)$$

$$1 - P(X \le 2) = 1 - \left[P(X=0) + P(X=1) + P(X=2)\right]$$

$$= 1 - \left[e^{-\lambda} + e^{-\lambda}\cdot\lambda + e^{-\lambda}\cdot\frac{\lambda^2}{2!}\right]$$

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$= 1 - \left[e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda} \right]$

Bernoulli and Binomial Random Variables

6. A satellite system consists of 4 components and functions adequately if at least 2 of the 4 components are working. If each component works independently with probability 0.6, what is the probability that the system functions adequately? *Source: Chapter 5, End-of-Chapter Problems, Problem 1, p. 212*

7. A communications channel transmits 0s and 1s, but a digit is incorrectly received with probability 0.2. To reduce error, "0" is sent as 00000 and "1" as 11111. The receiver uses majority decoding (decodes as "0" if at least 3 of the 5 received digits are 0). What is the probability that a message is incorrectly decoded? What independence assumption are you making? *Source: Chapter 5, End-of-Chapter Problems, Problem 2, p. 212*

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8. If each voter favors Proposition A with probability 0.7, what is the probability that exactly 7 of 10 voters favor this proposition? *Source: Chapter 5, End-of-Chapter Problems, Problem 3, p. 212*

9. Two hybrid parents (each carrying one dominant gene d and one recessive gene r) have 4 children. Each child independently has the outward (dominant) appearance with probability 3/4 and the recessive appearance with probability 1/4. What is the probability that exactly 3 of the 4 children have the outward appearance of the dominant gene? *Source: Chapter 5, End-of-Chapter Problems, Problem 4, p. 212*

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10. At least one-half of an airplane's engines are required to function for it to operate. If each engine independently functions with probability p, for what values of p is a 4-engine plane more likely to operate than a 2-engine plane? *Source: Chapter 5, End-of-Chapter Problems, Problem 5, p. 212*

11. If X and Y are binomial random variables with respective parameters (n, p) and $(n, 1-p)$, verify and explain the following identities: (a) $P\{X \leq i\} = P\{Y \geq n - i\}$; (b) $P\{X = k\} = P\{Y = n - k\}$. *Source: Chapter 5, End-of-Chapter Problems, Problem 7, p. 212*

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